

STANDARD OPERATING PROCEDURE

Coolant Concentration Control

Monitoring and Correction of Water-Miscible Metalworking Fluids

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Title	Coolant Concentration Control
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Document owner	S. Patel, Process Engineering
Approved by	K. Lindqvist, Maintenance Manager (2025-11-05)
Applies to	Greenfield Plant – Machining Cells 1–3 and central coolant system CS-1

1 Purpose

This procedure defines how the concentration and condition of water-miscible metalworking fluid (coolant) are monitored and corrected. Coolant kept within its specified limits protects tools and parts from corrosion, maintains surface finish and tool life, and reduces health risks such as dermatitis and bacterial contamination.

2 Scope

- 2.1

This procedure applies to all machines in Machining Cells 1–3 with individual coolant sumps and to the central coolant system CS-1.
- 2.2

It covers the semi-synthetic coolant NC-400, the only coolant approved for these machines. Neat cutting oils and grinding fluids are outside the scope.

3 References

Document	Title
SOP-00123	Machine Maintenance SOP
SDS NC-400	Safety Data Sheet, semi-synthetic coolant NC-400
TDS NC-400	Technical Data Sheet, semi-synthetic coolant NC-400
EHS-021	Metalworking Fluid Health Program
F-056-01	Coolant Log

4 Responsibilities

- Machine operator:** Checks the coolant level and appearance at the start of each shift and reports foam, odor or oil on the surface.
- Coolant technician:** Performs the measurements in this SOP, corrects the coolant and records all results on F-056-01.
- Maintenance Supervisor:** Reviews the Coolant Log weekly and arranges sump cleaning when required.
- EHS:** Runs the health program in EHS-021 and approves the use of biocides.

5 Safety / PPE

CAUTION

Coolant concentrate and biocides irritate the skin and eyes. Read the SDS before handling. Avoid skin contact with coolant and wash your hands before eating, drinking or smoking.

- 5.1** Nitrile gloves and safety glasses shall be worn when sampling, measuring or topping up coolant. A face shield must be worn when handling concentrate or biocide.
- 5.2** Biocides may only be added by trained coolant technicians and only in the dose approved by EHS.
- 5.3** Coolant spills must be cleaned up immediately; wet floors are a slip hazard.

6 Tools & Materials

Item	Specification
Refractometer	Handheld, 0–18 % Brix, automatic temperature compensation
Distilled water	For zeroing the refractometer
pH meter	Resolution 0.1 pH, calibrated weekly with pH 7.0 and pH 10.0 buffers
Dip slides	Combined bacteria/fungi slides; incubator at 30 °C
Sample bottles	Clean, 100 mL, labeled with machine number and date
Coolant proportioner	Set to 7 % premix, supplied with potable water
Tramp oil skimmer	Belt or disk type

7 Procedure

7.1 Sampling

- 7.1.1** Measure every sump weekly and after every recharge. Take samples with the coolant circulating for at least 15 minutes.
- 7.1.2** Draw the sample from the return flow near the pump inlet, not from the surface of the sump. Fill a clean sample bottle and label it.

7.2 Concentration measurement

- 7.2.1** Zero the refractometer with distilled water.
- 7.2.2** Place two or three drops of the sample on the prism, close the cover and read the Brix value at the boundary line.
- 7.2.3** Calculate the concentration: $\text{concentration (\%)} = \text{Brix reading} \times \text{refractometer factor}$. The refractometer factor of NC-400 is 1.2 (see TDS NC-400). Example: a reading of 5.8 Brix gives $5.8 \times 1.2 = 7.0 \%$.
- 7.2.4** The target concentration is 6–8 %. Record the Brix reading and the calculated concentration on F-056-01.

7.3 Concentration correction

Correct the concentration in accordance with Table 1. Never pour neat concentrate directly into a sump; always use premix from the proportioner or add water.

Table 1 – Correction of concentration

Measured concentration	Action
Below 5.0 %	Add 10 % premix; re-measure after 1 h; inform the supervisor
5.0–5.9 %	Add 10 % premix; re-measure at the next weekly check
6.0–8.0 %	No action; top up with 7 % premix only
8.1–9.0 %	Top up with water only
Above 9.0 %	Top up with water; check the proportioner setting

7.4 pH measurement

- 7.4.1 Measure the pH of the sample with the calibrated pH meter. The pH shall be 8.5-9.2.
- 7.4.2 If the pH is below 8.5, add pH buffer in accordance with TDS NC-400 and re-measure after 24 h.
- 7.4.3 A pH below 8.0 indicates strong bacterial activity: perform a dip slide test immediately (7.6) and inform the Maintenance Supervisor.

7.5 Tramp oil

- 7.5.1 Inspect the sump surface for tramp oil (leaked hydraulic oil and way oil). Tramp oil shall not exceed 2 % of the sump volume.
- 7.5.2 Remove tramp oil with the skimmer. Empty the skimmer collection container when it is three-quarters full.
- 7.5.3 Repeated high tramp oil indicates a leak: raise a maintenance work order to find the source (see SOP-00123).

7.6 Bacteria and fungi (dip slides)

- 7.6.1 Perform a dip slide test every 2 weeks, and whenever the coolant smells or the pH is below 8.0.
- 7.6.2 Immerse the dip slide in the sample for 10 s, drain it and incubate it at 30 °C for 48 h.
- 7.6.3 Compare the slide with the manufacturer's chart and record the result on F-056-01. Bacteria shall be below 100,000 CFU/mL; no fungal growth shall be visible.
- 7.6.4 At 100,000 CFU/mL or more, or with visible fungal growth, inform EHS; biocide may only be added as approved by EHS. At 1,000,000 CFU/mL or more, the sump shall be drained, cleaned and recharged.

7.7 Sump top-up and recharge

7.7.1 Top up sumps only with 7 % premix from the proportioner.

7.7.2 When a sump is recharged, drain it completely, clean it with the system cleaner specified in TDS NC-400, rinse it and fill it with 7 % premix. Measure the concentration after 15 minutes of circulation.

7.7.3 Record every top-up and recharge, with the volume added, on F-056-01.

8 Inspection & Acceptance Criteria

Parameter	Acceptance criterion	Frequency
Concentration	6–8 % (Brix × 1.2)	Weekly
pH	8.5–9.2	Weekly
Tramp oil	Max. 2 % of sump volume	Weekly
Bacteria (dip slide)	Below 100,000 CFU/mL	Every 2 weeks
Fungi (dip slide)	No visible growth	Every 2 weeks
Appearance and odor	No foam, no rancid odor	Each shift

8.1 Results outside these limits shall be corrected as described in Section 7 and the corrective action recorded.

8.2 A sump that does not meet the criteria after correction must be reported to the Maintenance Supervisor.

9 Records

9.1 All measurements, corrections, top-ups and recharges are recorded on the Coolant Log F-056-01, one log per sump.

9.2 Coolant logs shall be retained for 3 years.

10 Revision History

Rev.	Date	Author	Description of change
A.1	2019-03-11	S. Patel	Initial release.
B.1	2022-08-29	S. Patel	pH range and dip slide testing added.
C.1	2025-11-05	S. Patel	Refractometer factor for NC-400 updated to 1.2; tramp oil limit reduced to 2 %. Released.